Experiment -1

AIM - Introduction to GNU Sim8085 Microprocessor Simulator.

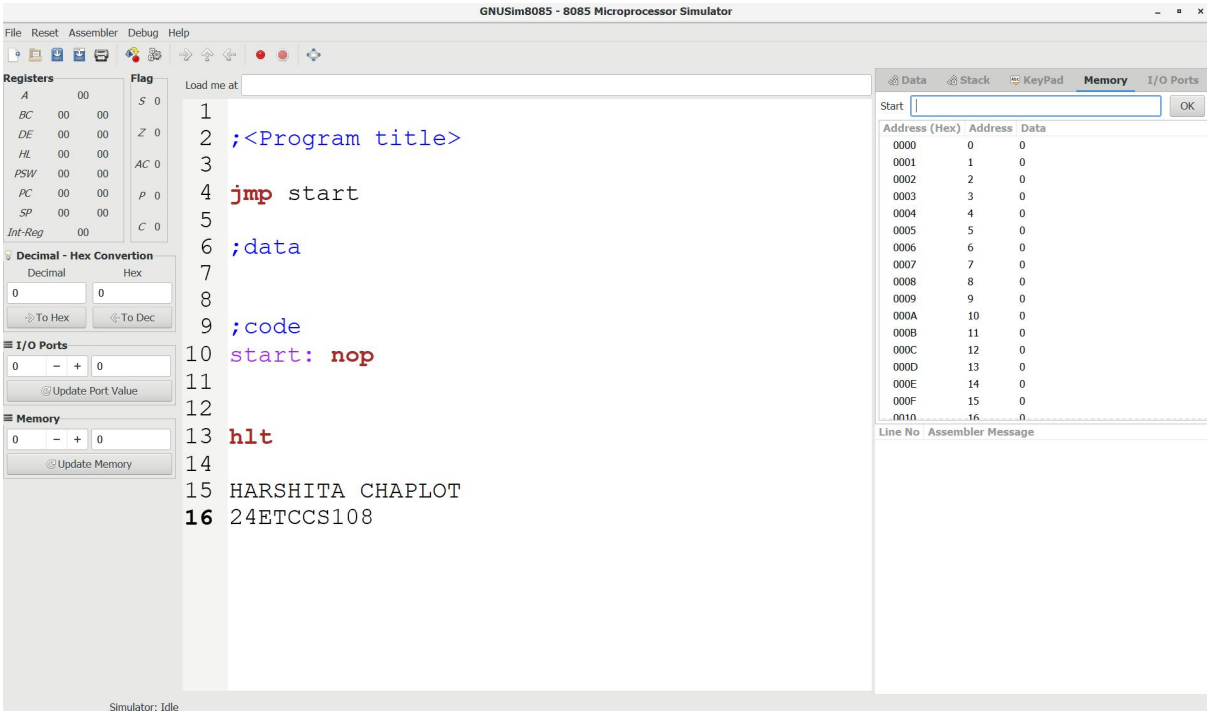
TOOL - GNU Sim8085

THEORY - GNUSim8085 is a graphical simulator designed for the Intel 8085. It allows users to write, assemble, and execute assembly language programs and helps in understanding how instructions are executed in the microprocessor.

On the left side, the simulator shows the register section, including registers A, B, C, D, E, H, and L along with the Program Counter (PC), Stack Pointer (SP), and flag status.

The middle section contains the program editor or workspace where the assembly program is written and edited.

On the right side, the simulator displays the instruction set and memory options, which help the user observe instructions and memory during program execution.



RESULT – Hence, the GNU SIM8085 Simulator was executed successfully.

Experiment -2

AIM - Introduction to GNU Sim8085 Microprocessor Simulator.

TOOL - GNU Sim8085

THEORY - In this experiment, a simple 8085 assembly program is executed to swap (interchange) the data stored at two memory locations. The microprocessor cannot directly swap memory locations, so it uses registers as temporary holders.

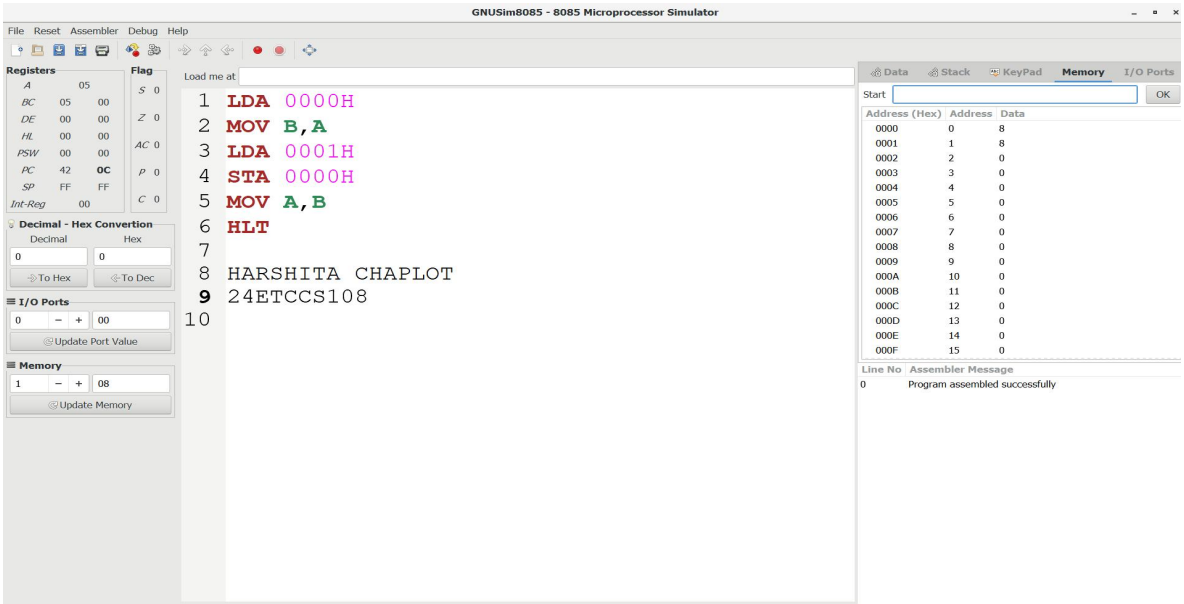
First, the data from the first memory location is loaded into the accumulator. Then, it is moved to register B for temporary storage. Next, the data from the second memory location is loaded into the accumulator and stored back into the first memory location. Finally, the data stored in register B is moved back to the accumulator.

The instructions used include:

- LDA – Load accumulator from memory , MOV – Transfer data between registers , STA– Store accumulator to memory , HLT – Terminate program execution

This experiment helps in understanding data transfer operations and the concept of using registers as intermediaries.

PROGRAM LOGIC (flow in words):
Memory[0000H] → A → B
Memory[0001H] → A → Memory[0000H]
B → A



RESULT – Hence, the 8085 assembly program to interchange the contents of two memory locations was executed successfully using the GNU Sim8085 Simulator.

Experiment -3

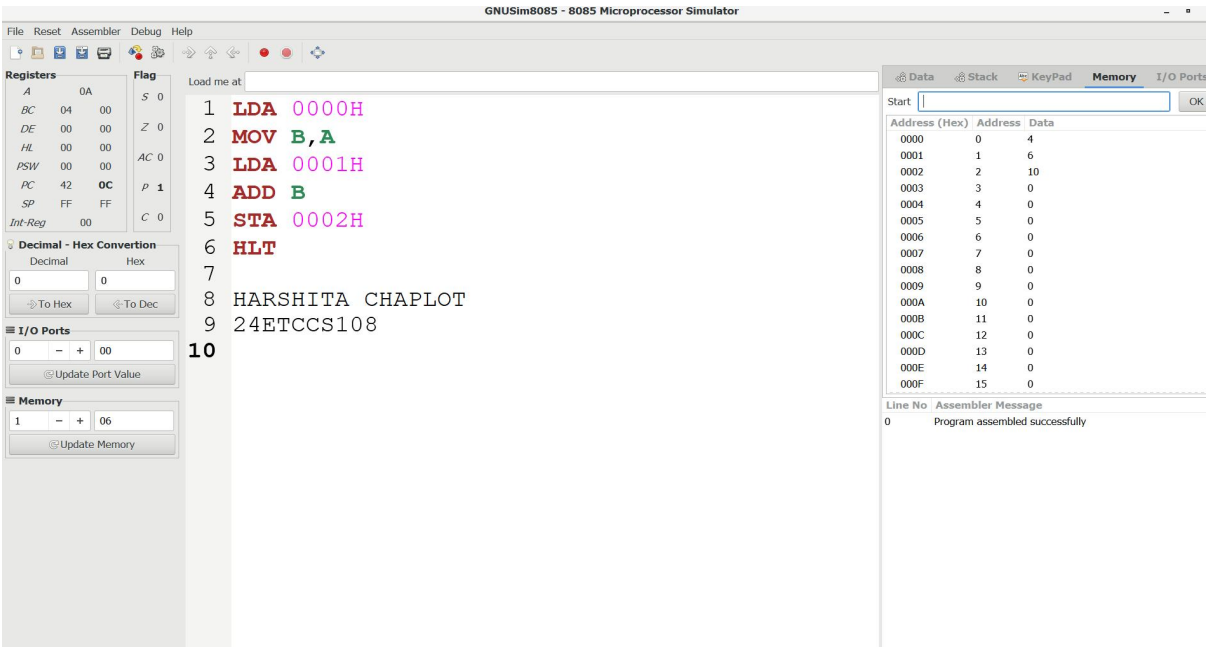
AIM – To perform 8-bit addition GNU Sim8085 Microprocessor Simulator.

TOOL - GNU Sim8085

THEORY - This program adds two numbers stored in memory. The first value (04H) is stored at address 0000H and the second value (06H) at address 0001H. One value is moved to a register, then added with the other, and the result is stored in memory location 0002H.

Instructions used:
LDA – Load accumulator
MOV – Transfer data
ADD – Add register to accumulator
STA – Store result
HLT – Stop execution

PROGRAM LOGIC:
Memory[0000H] → A → B
Memory[0001H] → A
A + B → A
A → Memory[0002H]



RESULT –

The sum of 04H and 06H is 0AH, which is stored at memory location 0002H successfully using GNU Sim8085.

EXPERIMENT – 4

AIM – To add two 16-bit numbers using 8085 microprocessor.

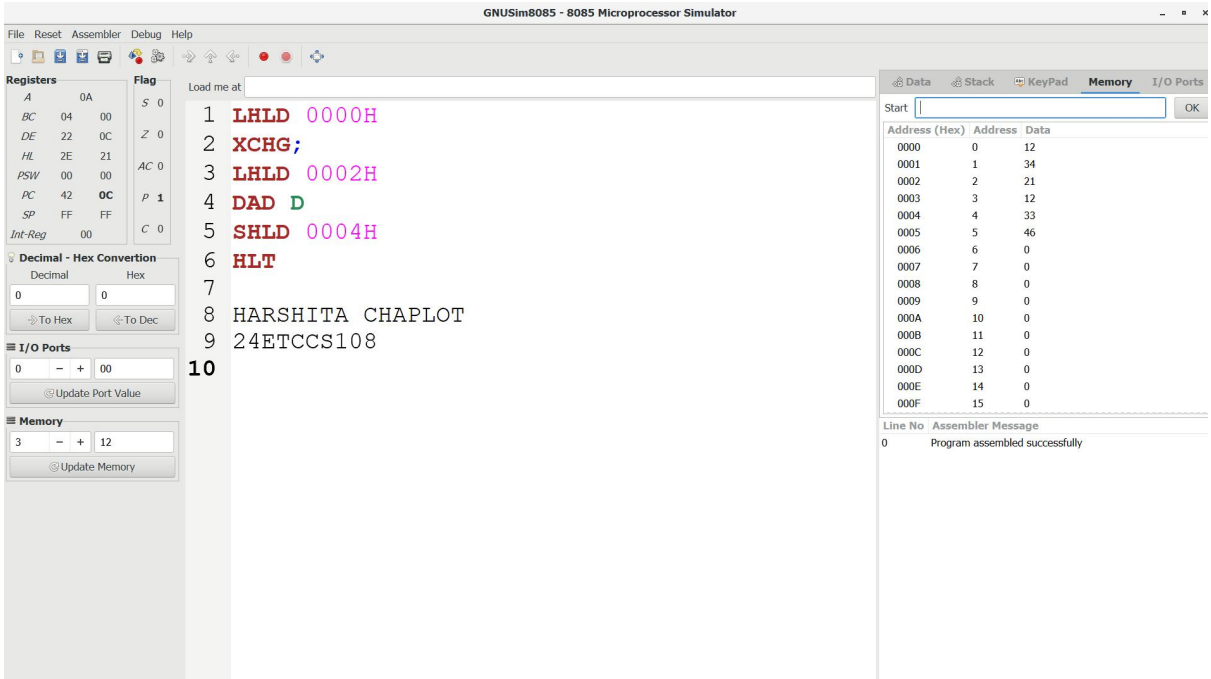
TOOL – GNU Sim8085

THEORY – This program adds two 16-bit numbers stored in memory using register pairs. The first number (stored at 0000H–0001H) is loaded into HL, then exchanged with DE. The second number (stored at 0002H–0003H) is loaded into HL and added with DE. The result is stored in memory.

PROGRAM LOGIC WITH VALUES**:**
(0001H–0000H) = 34H 12H → First number = 3412H
(0003H–0002H) = 12H 21H → Second number = 1221H

3412H + 1221H = 4633H

Result → Memory[0004H] = 33H
Memory[0005H] = 46H



RESULT –
The sum of two 16-bit numbers (3412H and 1221H) is 4633H, stored successfully at memory locations 0004H and 0005H.

EXPERIMENT – 5

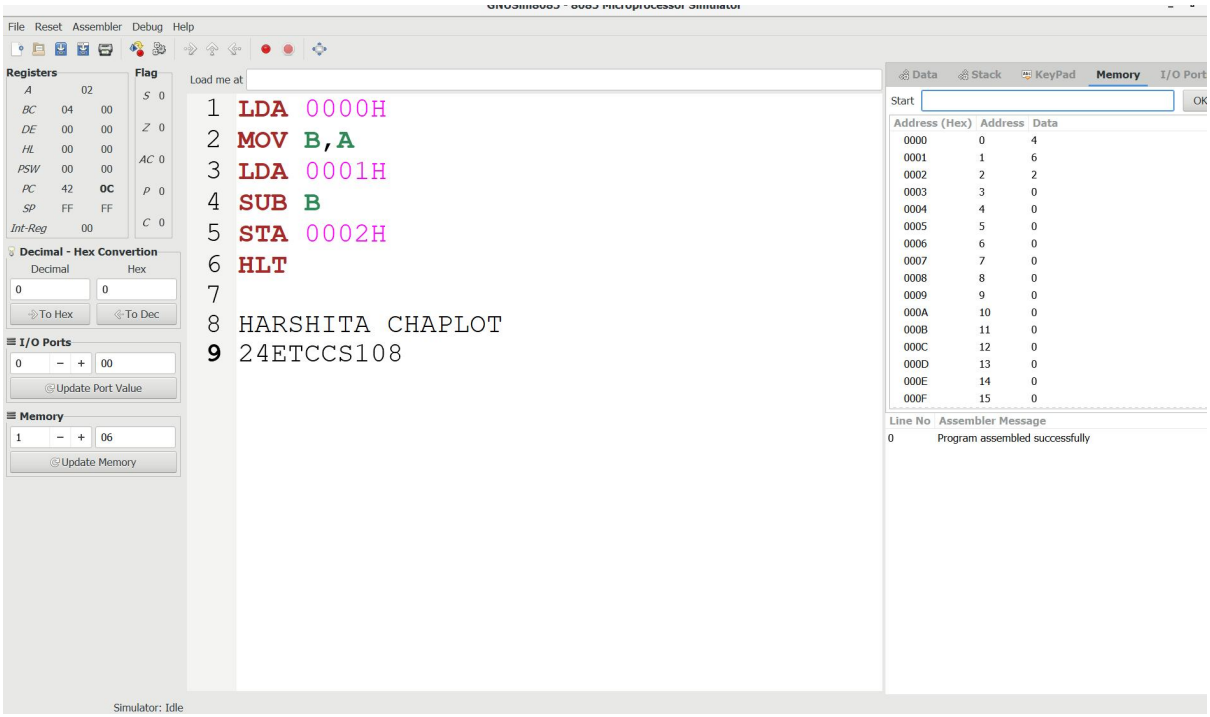
AIM – To write and execute an 8085 program to subtract two numbers.

TOOL – GNU Sim8085

THEORY – This program performs subtraction of two numbers stored in memory using the accumulator. The first value is stored in a register, the second value is loaded, and subtraction is performed. The result is then stored in memory.

PROGRAM LOGIC WITH VALUES:
04H → A → B
06H → A
A − B = 02H

02H → Memory[0002H]



RESULT –
The result of subtraction is 02H, stored successfully at memory location 0002H.

EXPERIMENT – 6

AIM – To write and execute an 8085 program to subtract two 16-bit numbers.

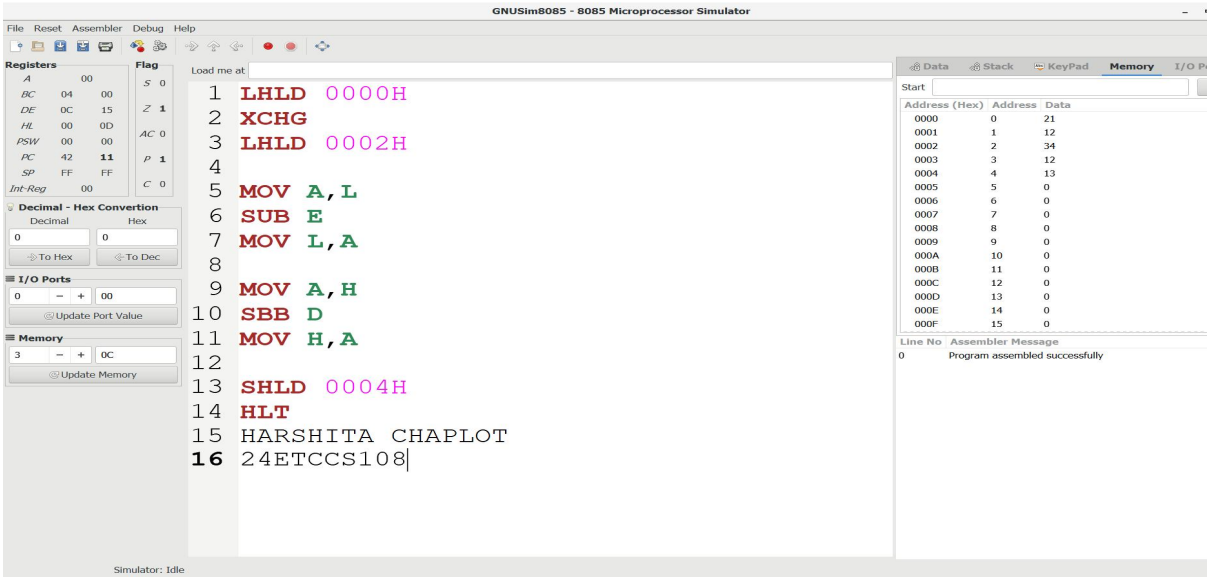
TOOL – GNU Sim8085

THEORY – This program performs subtraction of two 16-bit numbers using register pairs. Since 8085 has no direct 16-bit subtraction instruction, subtraction is done in two steps: first the lower bytes, then the higher bytes with borrow. The result is stored back in memory.

COMMANDS USED (8085 Instructions):
LHLD – Load HL pair direct
XCHG – Exchange HL and DE
SUB – Subtract register from A
SBB – Subtract with borrow
STA – Store accumulator
SHLD – Store HL pair direct
HLT – Stop execution

PROGRAM LOGIC WITH VALUES (example):
First number = 3412H
Second number = 1221H

3412H − 1221H = 21F1H
Memory[0004H] = F1H
Memory[0005H] = 21H



RESULT –
The subtraction of two 16-bit numbers was performed successfully, and the result was stored in memory.